Fiscal Year:	FY 2017	**Task Last Updated:**	FY 06/20/2017
PI Name:	Blaber, Elizabeth Ph.D.		
Project Title:	Osteogenic Differentiation of Somatic Stem Cells in Space: A Study Investigating the Role of CDKN1a/p21 on Mesenchymal Stem Cell Proliferation, Differentiation, and Regeneration in Microgravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
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Zip Code:	12180	**Congressional District:**	20
Comments:	NOTE: Moved to Rensselaer Polytechnic Institute in fall 2019 from NASA Ames Research Center.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	11/01/2014	**End Date:**	07/15/2019
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 7/15/2019 as PI moved to Rensselaer Polytechnic Institute and a new grant 80NSSC20K1125 was issued for continuation at RPI (Ed., 8/27/20) NOTE: End date changed to 12/31/2020 per F. Hernandez/ARC (Ed., 6/20/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Almeida, Eduardo Ph.D. (NASA Ames Research Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Spaceflight factors, including microgravity and space radiation, are known to affect somatic stem cell-based tissue regeneration and cause significant degenerative conditions in multiple mammalian tissues, posing a significant risk for long-term human space exploration. Our previous spaceflight research has shown an inability of both embryonic and somatic stem cells to proliferate, differentiate, and regenerate damaged cells in microgravity conditions, which may be the cause of the widespread tissue degeneration observed in space. Our laboratory has recently published key evidence that spaceflight induces over expression of CDKN1a/p21, a master cell cycle and tissue regeneration inhibitor, in proliferating osteoprogenitor cells, potentially providing a molecular mechanism for spaceflight tissue regenerative deficits. In this proposal, we seek to test the hypothesis that the key molecular mechanism causing the degenerative effects of microgravity on adult stem cells is activation of cell cycle arrest and impairment of differentiation via CDKN1a/p21 overexpression. We specifically hypothesize that in microgravity, normal mouse bone marrow mesenchymal stem cells with the CDKN1a/p21 gene will be inhibited from proliferating and differentiating normally relative to 1g, but that mesenchymal stem cells from CDKN1a/p21-null mice will show unrestrained proliferation and differentiation that is not sensitive to microgravity conditions. To test this hypothesis, we are proposing a spaceflight experiment to Section B.2 ‘Mammalian Cell, Tissue and Organ Generation and Degeneration in Space’ of the NASA Research Announcement NNH14ZTT001N. We specifically propose to conduct an experiment in the NASA Bioculture System Hardware in which the stromal stem cell population isolated from bone marrow of CDKN1a/p21-null and wildtype mice will be taken to the International Space Station (ISS) National Laboratory frozen on the surface of collagen-coated alginate microcarrier beads, injected into bioreactors, and cultured in osteogenic medium to induce differentiation and mineralized bone nodule formation in a 30 day culture. The successful completion of this experiment will conclusively determine the role of CDKN1a/p21 in stem cell regenerative health in space. Specifically, if we observe inhibition of osteogenesis by microgravity in wildtype stem cells, but not in CDKN1a/p21-null cells, this will support our hypothesis that the induction of cell cycle arrest by microgravity in adult stem cells is a key molecular mechanism causing the observed degenerative effects of space on mammalian tissues. This will provide a new perspective on the problem of stem cell regenerative health and medicine in space as well as on Earth, and will provide fundamental insight into stem cell regenerative mechanisms that will aid our understanding of basic stem cell functions on Earth and in space. This research addresses NASA’s and the National Research Council's (NRC’s) Decadal Survey call to study the molecular and cellular mechanisms of bone loss and to conduct basic research into the mechanisms of bone adaptation to microgravity using transgenic animal models. It also will address the call to understand the precise signaling mechanisms responsible for reduced bone formation in space. This research is highly significant, as it proposes to investigate a molecular mechanism, based on previous spaceflight results, that has the potential to explain and mitigate a broad range of microgravity induced tissue degenerative health conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The successful completion of this experiment will conclusively determine the role of CDKN1a/p21 in stem cell regenerative health in space. Specifically, if we observe inhibition of osteogenesis by microgravity in wildtype stem cells, but not in CDKN1a/p21-null cells, this will support our hypothesis that the induction of cell cycle arrest by microgravity in adult stem cells is a key molecular mechanism causing the observed degenerative effects of space on mammalian tissues. This will provide a new perspective on the problem of stem cell regenerative health and medicine in space as well as on Earth, and will provide fundamental insight into stem cell regenerative mechanisms that will aid our understanding of basic stem cell functions on Earth and in space.
Task Progress:	Due to delays in spaceflight opportunities to conduct the research outlines in the NASA Grant, ‘Osteogenic Differentiation of Stem Cells in Space: A Study Investigating the role of CDKN1a/p21 on Mesenchymal Stem Cell Proliferation, Differentiation, and Regeneration in Microgravity,’ we proposed an expansion of our existing methodology to include ground-based studies, whilst still achieving the same science aims. This request was approved in early 2016 and provides the basis for the current report. Specifically, we restructured the period of grant performance from 3 years to two separate 2 year phases whilst still maintaining the same costs. This restructuring included an initial 2-year ground research and definition phase beginning in FY16 and continuing to FY18 and a subsequent 2 year spaceflight phase beginning when a flight opportunity becomes available. In the initial ground research and definition phase beginning in March 2016, we began addressing Specific Aims 1A, 1B, and 1C of our original proposal. These aims included rigorous ground based tests to characterize the role of CDKN1a/p21 of osteoprogenitor differentiation and to optimize the exact experimental procedures that would be used for flight. The analysis of these mechanisms was already proposed in a time-course differentiation experiment using primary bone marrow stem cells from long bones of wildtype and CDKN1a/p21 -/- knockout (KO) mice induced to differentiate using specific osteogenic factors. To further assess the regenerative capacity of KO mice, we expanded these studies to include analysis of differentiation studies using osteoprogenitors isolated from CDKN1a/p21 KO mice and WT (wild type) in standard culture conditions and on alginate beads to determine gene expression baselines that can then be used for flight, an aging study to assess the effects of age on regenerative processes in both WT and KO mice, and a loading experiment to assess the effect of increased load on regenerative capacity of these cells. Due to the long lead time and high cost of KO mice, we began full scale breeding colonies of both WT and KO mice in early 2016 in order to fulfill the demands for mouse cells from these experiments. We are currently assessing bones and differentiation data from 4 week, 8 week, and 16 week old WT and KO mice and are aging mice to assess their regenerative capacity at 6 months, 12 months, and 18 months old. Furthermore, we have collected bone marrow stem cells from mice hindlimb unloaded for 15 and 30 days and are assessing their differentiation capacity in vitro. These samples have already been collected for male mice and the female mouse experiment is being conducted in early 2017. We have also initiated analysis of gene expression measured by Affymetrix gene arrays using RNA extracted from bone marrow samples and in a time course of differentiations for specific analysis of the CDKN1a/p21 pathway and potential involvement of the CIRBP gene. Gene expression analysis and processing underway includes data normalization, grouping, and qc for significance and gene ontology analysis using GeneSpring software. In addition, analysis of alterations in gene expression pathway is also being conducted using KEGG and GSEA. Finally, we have initiated in-vitro studies of cell cultures subjected to uniaxial loading and also started to investigate gap junction and integrin adhesion signaling in the context of p21 status, and how it may contribute to sensing and responding to gravity mechanotransduction. We are planning to move into optimization experiments in early 2017 in preparation for a flight opportunity as well as continuing our in-vitro mechanical loading experiments and our differentiation studies. Optimization experiments will be conducted in cell culture dishes, in the bioreactors attached to a CellMax system and in the Bioculture System Hardware

	itself. We are also moving into an optimization phase with WL-2 personnel to determine sampling and isolation procedures using the WL-2 suite of equipment.
Bibliography Type:	Description: (Last Updated: 04/03/2025)
Abstracts for Journals and Proceedings	Juran CM, Blaber EA, Almeida EAC. "Mechanotransductive regulation of gap-junction activity between MLO-Y4 osteocyte-like and MC3T3-E1 osteoblast-like cells in Three-Dimensional Co-Culture." Presented at 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. , Oct-2016
Abstracts for Journals and Proceedings	Blaber EA, Almeida EAC. "The role of the CDKN1a/p21 pathway in microgravity-induced arrest of stem cell-based tissue regeneration." Presented at International Society for Gravitational Physiology (ISGP), European Space Agency (ESA), Centre National d'Etudes Spatiales (CNES) Joint Life Science Meeting ‘Life in Space for Life on Earth,’ Toulouse, France, June 5-10, 2016. International Society for Gravitational Physiology (ISGP), European Space Agency (ESA), Centre National d'Etudes Spatiales (CNES) Joint Life Science Meeting ‘Life in Space for Life on Earth,’ Toulouse, France, June 5-10, 2016. , Jun-2016
Abstracts for Journals and Proceedings	Blaber EA, Shirazi-Fard Y, Almeida EAC. "The role of the CDKN1a/p21 pathway in microgravity-induced arrest of stem cell-based tissue regeneration." Presented at 2016 NASA Human Research Program Investigators’ Workshop, Galveston, TX, February 8-11, 2016. 2016 NASA Human Research Program Investigators’ Workshop, Galveston, TX, February 8-11, 2016. , Feb-2016